

UNIT 3

Q1: DSDV Protocol (Very Easy)

👉 Idea: Every node already knows all routes (like Google Maps saved offline)

◆ How it works:

- Each node keeps a routing table (destination, next hop, distance)
- Uses sequence number → to know latest route

◆ Connection Establishment:

- No searching needed ❌
- Just check table ✔
- Route is ready → instant connection

◆ Data Transfer:

A → B → C → D (hop by hop)

◆ Memory Trick:

👉 "DSDV = Direct, Stored, Directly Used"

🦉 Q2: MAC Layer (Adhoc)

👉 Idea: Controls who uses wireless channel

◆ Problems:

- Hidden → collision
- Exposed → unnecessary waiting
- Low battery 🔋
- Limited bandwidth

◆ Goals:

- Fair usage
- Less delay
- Save energy

◆ Types:

- Contention (fight for channel)
- Contention-free (fixed turn)
- Hybrid (mix)

👉 Memory Trick:

👉 "MAC = Manage Access Carefully"

🌀 Q3: Routing Protocol Types

👉 Idea: How route is found

◆ 3 Types:

- Proactive: route always ready (DSDV)
- Reactive: find when needed (AODV, DSR)
- Hybrid: mix

👉 Memory Trick:

👉 "Pro = Ready, Re = Request, Hy = Hybrid"

🌐 Q4: Clustered Sensor Network

👉 Idea: Divide network into groups

- Sensor nodes → collect data
- Cluster Head → manage group
- Base Station → final receiver

◆ Flow:

Nodes → CH → Base Station

👉 Memory Trick:

👉 "Cluster = Group + Leader"

📊 Q5: MACAW Protocol

👉 Idea: Improved collision control

◆ Steps:

RTS → CTS → DS → DATA → ACK

👉 Meaning:

- Ask → Allow → Inform → Send → Confirm

👉 Memory Trick:

👉 "5 Steps = Safe Communication"

🌱 Q6: Sensor Network Protocols

◆ Types:

- Flat → all equal
- Hierarchical → clusters
- Location → GPS-based

👉 Memory Trick:

👉 "Flat, Cluster, Location"

💻 Q7: AODV vs DSR

◆ AODV:

- Uses routing table
- Finds route when needed

◆ DSR:

- Full path inside packet

👉 Difference Trick:

👉 "AODV = Table, DSR = Full Route"

⚠️ Q8: Routing Design Issues

👉 Problems while designing:

- Nodes move
- Low battery
- Limited bandwidth
- Security issues

👉 Memory Trick:

👉 "Move + Energy + Bandwidth + Security"

Q9: MAC Design Issues & Goals

👉 Same as Q2 but important:

Issues:

- Hidden / Exposed
- Energy
- Bandwidth

Goals:

- Efficiency
- Fairness
- Low delay

👉 Memory Trick:

👉 "Problem → Fix → Goal"

Q10: Adhoc vs Sensor Network

Feature	Adhoc	Sensor
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Purpose	Communication	Sensing
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Mobility	High	Low
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Energy	Medium	Very low
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👉 Memory Trick:

👉 "Adhoc = Talk, Sensor = Sense"

Q11: DSR Protocol

👉 Idea: Full route stored in packet

◆ Route Discovery:

A → B → C → D

◆ Route Maintenance:

- Link breaks → send error
- Find new route

👉 Memory Trick:

👉 "DSR = Direct Stored Route"

Q12: Layered Architecture (WSN)

👉 Like OSI model:

- Physical → signals
- Data link → error
- Network → routing
- Transport → reliability
- Application → user

👉 Memory Trick:

👉 "P D N T A"

Q13: Infrastructure vs Infrastructure-less

Infra No Infra

Fixed base No base

Central control Distributed

👉 Memory Trick:

👉 "Infra = Fixed, Adhoc = Flexible"

Q14: Hidden & Exposed Problem

Hidden:

A & C → B → collision

Exposed:

C stops even when safe

👉 Memory Trick:

👉 "Hidden = Collision, Exposed = Waste"

🌟 Q15: DDoS Attack

👉 Many computers attack one server

Steps:

Botnet → Command → Flood → Crash

👉 Memory Trick:

👉 "Many attack → System down"

🌐 Q16: WSN + LEACH Protocol

◆ Types of WSN:

- Flat
- Cluster
- Location

◆ LEACH Phases:

1. Setup:

- Select Cluster Head
- Form clusters
- Create schedule

2. Steady:

- Nodes send data
- CH aggregates
- Send to base

👉 Memory Trick:

👉 "Setup → Send"

🚀 FINAL SUPER SHORT REVISION

👉 DSDV = Table ready

👉 MAC = Channel control

👉 Routing = Pro, Re, Hybrid

👉 Cluster = Group + CH

- 👉 MACAW = 5 steps
- 👉 Sensor = Flat / Cluster / Location
- 👉 AODV vs DSR = Table vs Path
- 👉 Issues = Mobility + Energy
- 👉 Adhoc vs Sensor = Talk vs Sense
- 👉 DSR = Full route
- 👉 Layers = PDNTA
- 👉 Infra = Fixed
- 👉 Hidden = Collision
- 👉 DDoS = Many attack
- 👉 LEACH = Setup + Send

UNIT 4

🧠 Q1 & Q9: CBC Mode (Cipher Block Chaining)

👉 **Idea:** Each block depends on previous block (chain)

- First block uses **IV (starting value)**
- Next block uses **previous ciphertext**
- So even same data → different output

👉 **Formula:**

$$C_i = E(P_i \oplus C_{i-1})$$

👉 **Easy Memory Trick:**

👉 **“CBC = Chain Before Cipher”**

👉 **Important Points:**

- Secure ✓
 - Slow ✗ (no parallel)
 - Error spreads to next block
-

📖 Q2: Substitution Cipher

👉 **Idea:** Replace letters with other letters

- **Monoalphabetic:** Same replacement always
→ A → D (fixed) ✗ weak
- **Polyalphabetic:** Changes with position
→ More secure ✓

👉 **Example:**

HELLO → KHOOR (mono)

👉 **Memory Trick:**

👉 **“Mono = One rule, Poly = Many rules”**

⚠️ Q3: Attacks (Integrity & Availability)

👉 **Integrity = Data should not change**

👉 **Availability = System should work always**

Integrity Attacks:

- Modify message
- Fake message
- Replay old message

Availability Attacks:

- DoS / DDoS → overload
- Jamming → block signals

Memory Trick:

 “Integrity = Change, Availability = Stop”

Q4: Network Attack

 **Idea:** Attacker tries to break system

Passive:

- Only listens 
- No change
- Hard to detect

Active:

- Changes data 
- Easy to detect

Memory Trick:

 “Passive = Watch, Active = Attack”

Q5: Need for Security

 Why security needed?

- Protect data 
- Keep system running 
- Build trust 
- Avoid loss 

Memory Trick:

 “Security = Protect + Trust + Work”

Q6: Block Cipher Modes

👉 Different ways to encrypt blocks

- **ECB:** Simple but weak ❌
- **CBC:** Secure ✓
- **CFB:** Like stream ✓
- **OFB:** No error spread ✓

👉 **Memory Trick:**

👉 “ECB bad, CBC best, CFB flow, OFB smooth”

Q7: DDoS Attack

👉 **Idea:** Many computers attack one server

- Bots send too many requests
- Server crashes

👉 **Flow:**

Botnet → Attack → Overload → Down

👉 **Memory Trick:**

👉 “Many attack = System down”

Q8 & Q14: Stream Cipher

👉 **Idea:** Encrypt bit-by-bit (not block)

👉 **Formula:**

$$C = P \oplus K$$

- Uses **keystream (random bits)**
- Very fast ⚡

👉 **Problem:**

If key reused → unsafe ❌

👉 **Memory Trick:**

👉 “Stream = Continuous XOR”

Q10: Security Threats

Unauthorized Access:

- Login without permission

MITM:

- Attacker sits in between

👉 Example:

User → Attacker → Server

👉 **Memory Trick:**

👉 “MITM = Middle Man”

Q11: Security Principles

👉 Core rules of security:

- Confidentiality → keep secret
- Integrity → no change
- Authentication → who are you
- Authorization → what you can do

👉 **Memory Trick:**

👉 “CIA + Auth”

Q12: ECB & CFB

ECB:

- No connection between blocks
- Same input → same output ❌

CFB:

- Uses previous ciphertext
- More secure ✓

👉 **Memory Trick:**

👉 “ECB = Same, CFB = Chain”

Q13: CFB Process

👉 **Idea:** Uses previous output

- IV → Encrypt → XOR → Output
- Output used again

👉 **Memory Trick:**

👉 “CFB = Feedback loop”

⚡ **Q15: Security Principles & Services**

👉 Same as Q11 but with **implementation**

- Principle = Goal
- Service = Method

👉 Example:

Integrity → Hashing

👉 **Memory Trick:**

👉 “Principle = What, Service = How”

🔑 **Q16: Symmetric vs Asymmetric**

Symmetric:

- One key
- Fast
- Used for data

Asymmetric:

- Two keys
- Secure
- Used for key sharing

👉 **Memory Trick:**

👉 “Symmetric = Speed, Asymmetric = Security”

🚀 **FINAL SUPER SHORT REVISION (1 MIN)**

👉 CBC = Chain

👉 Substitution = Replace letters

👉 Attacks = Change or Stop

- 👉 Passive = Listen, Active = Modify
- 👉 Security = Protect data
- 👉 Modes = ECB bad, CBC good
- 👉 DDoS = Many attack one
- 👉 Stream = XOR
- 👉 MITM = Middle attacker
- 👉 CIA = Security base
- 👉 ECB = Weak
- 👉 CFB = Feedback
- 👉 Crypto = Symmetric fast, Asymmetric secure

UNIT 5

Q1: PKIX Model (Very Easy Explanation)

- **Meaning (Simple)**

- PKIX = Public Key Infrastructure using X.509
 - It is a **system to manage digital certificates** (like digital ID cards).
 - Used in HTTPS, email security, VPN, etc.
-

- **Purpose**

- Authentication → Check “who you are”
 - Trust → Build trust between users
 - Security → Enable encryption & digital signatures
-

- **Components (Easy Trick: E-C-R-R-R)**

1. **End Entity** → User (browser, email user)
 2. **CA (Certificate Authority)** → Issues certificate (like passport office)
 3. **RA (Registration Authority)** → Verifies identity
 4. **Repository** → Stores certificates
 5. **Revocation System** → Removes invalid certificates (CRL/OCSP)
-

- **Working (Step-by-Step)**

1. User requests certificate
 2. RA verifies identity
 3. CA issues certificate
 4. Certificate is stored/shared
 5. Receiver verifies using CA
 6. If unsafe → certificate revoked
-

- **Conclusion**

- PKIX = **Trust System for Internet Security**

🔒 Q2: AES (Advanced Encryption Standard)

- **Meaning**

- AES is a **modern symmetric encryption algorithm**
 - Replaced DES because it is **more secure**
-

- **Key Features**

- Block size = 128 bits
 - Key sizes = 128 / 192 / 256 bits
 - Uses multiple rounds (10, 12, 14)
-

- **AES Process (Easy Flow)**

👉 **Key Expansion → AddRoundKey → Rounds → Final Round**

- **Main Steps in Each Round**

1. **SubBytes** → Replace bytes
 2. **ShiftRows** → Shift rows
 3. **MixColumns** → Mix data
 4. **AddRoundKey** → XOR with key
-

- **Short Diagram Idea (Draw in exam)**

Plaintext → SubBytes → ShiftRows → MixColumns → AddRoundKey → Ciphertext

- **Conclusion**

- AES = **Very Secure + Fast + Widely Used**
-

🔒 Q3: Hash, Digital Signature, Digital Certificate

1) Hash Function

- Converts data → fixed-size value
- Used for **Integrity (data not changed)**

✓ Properties

- One-way (cannot reverse)
- Small change → big output change

✓ Example: SHA-256

2) Digital Signature

- Proves **who sent message + message not changed**

Working

Sender:

- Message → Hash → Encrypt with private key

Receiver:

- Decrypt using public key
- Compare hash

✓ Provides:

- Authentication
 - Integrity
 - Non-repudiation
-

3) Digital Certificate

- Digital ID issued by CA

✓ Contains:

- Name
- Public key
- CA signature

✓ Use:

- HTTPS ( symbol in browser)
-

• Conclusion

- Hash = Integrity
 - Signature = Trust
 - Certificate = Identity
-

Q4: DES Algorithm (Easy)

- **Meaning**
 - Old symmetric encryption algorithm
 - Uses **64-bit data + 56-bit key**
-

- **Steps**
 1. Initial Permutation
 2. Split → L0 & R0
 3. 16 rounds
 4. Swap
 5. Final Permutation
-

- **Core Formula**

 (This is most important for exam)

$$L_i = R_{i-1}, R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

- **Conclusion**
 - DES = Old + Less secure → replaced by AES
-

Q5: Symmetric Key Encryption + DES

- **Definition**
 - Same key used for encryption & decryption
-

- **Flow**

Plaintext → Encrypt → Ciphertext → Decrypt → Plaintext

- **Examples**

- DES
- AES

- **Conclusion**

- Fast but **key sharing is difficult**

 Q6: RSA Algorithm (Very Easy)

- **Meaning**

- Asymmetric encryption (2 keys)

- **Keys**

- Public key → Encrypt
- Private key → Decrypt

- **Formula**

$$C = M^e \bmod n$$

$$M = C^d \bmod n$$

- **Conclusion**

- Secure but slower than symmetric

 Q7: Private Key + Diffie-Hellman

- **Private Key**

- Secret key (only owner knows)
- Used for decryption & signing

- **Diffie-Hellman (Easy Idea)**

- Create **shared key without sending it**
-

- **Formula**

$$K = \alpha^{AB} \bmod q$$

- **Conclusion**

- Secure key exchange method
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Q9: Groups, Rings, Prime Numbers

- **Group**

- Set with one operation
- Has identity & inverse

Example: $(\mathbb{Z}, +)$

- **Ring**

- Set with 2 operations $(+, \times)$

Example: Integers

- **Prime Numbers**

- Only divisible by 1 and itself

✓ Used in RSA

- **Conclusion**

- These are **math base of cryptography**
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Q10: Security Mechanisms + Encryption/Decryption

- **Security Mechanisms**

- Encryption
 - Authentication
 - Access Control
 - Digital Signature
 - Firewall
 - IDS
 - VPN
-

- **Encryption**

- Plaintext → Ciphertext
-

- **Decryption**

- Ciphertext → Plaintext
-

- **Conclusion**

- Encryption = Lock
 - Decryption = Unlock
-

 Q11: DES Working (Short Revision)

- 64-bit data
 - 16 rounds
 - Feistel structure
 - Same process for decryption (reverse keys)
-

 Q12: Digital Signature Standard (DSS)

- **Generation**

1. Message → Hash
 2. Encrypt using private key
 3. Signature created
-

- **Verification**

1. Hash message
2. Decrypt signature
3. Compare

✓ Same → Valid

✓ Different → Invalid

- **Conclusion**

- DSS ensures **authentic + secure communication**

 Final Exam Trick

 Remember this shortcut:

- **PKIX → Certificates system**
- **AES → Modern encryption**
- **DES → Old encryption**
- **RSA → Public key**
- **Hash → Integrity**
- **Signature → Authentication**
- **Diffie → Key exchange**

UNIT 6

✓ Q1 (Extra Explanation): Hardware Attack vs Software Attack + MITM

◆ Difference (Improved Explanation – 8–10 Marks)

No.	Hardware Attack	Software Attack
1	Targets physical components of a system	Targets software/programs
2	Examples: CPU, RAM, keyboard, hard disk	Examples: OS, apps, browsers
3	Requires physical access (mostly)	Can be done remotely via internet
4	Involves tampering or physical damage	Involves code/data manipulation
5	Difficult to detect using tools	Easier detection (antivirus, logs)
6	Example: Hardware keylogger	Example: Virus, ransomware
7	Can cause permanent hardware damage	Mostly affects data/software
8	Prevented using physical security (locks, CCTV)	Prevented using firewalls, antivirus
9	Less common but severe	Very common in cyber attacks
10	Needs direct contact with system	No physical presence needed

✚ Simple Difference:

- Hardware attack → Physical damage
- Software attack → Digital damage

◆ Man-in-the-Middle (MITM) Attack (Improved Note)

✚ Definition

A **MITM attack** is a cyber attack where an attacker secretly intercepts communication between two users and may modify the data.

Explanation

- Attacker sits **between sender and receiver**
 - Both think they are communicating directly
 - But all data passes through attacker
-

Diagram

Sender → Attacker → Receiver

Working Steps

1. Sender sends data
 2. Attacker intercepts
 3. Attacker reads/modifies data
 4. Sends it to receiver
 5. Receiver remains unaware
-

Types

- IP Spoofing
 - DNS Spoofing
 - Session Hijacking
 - Wi-Fi Eavesdropping
 - Email Hijacking
-

Effects

- Password theft
- Data manipulation
- Identity theft
- Financial loss

Prevention

- Use HTTPS (SSL/TLS)
- Avoid public Wi-Fi
- Use VPN
- Enable MFA
- Verify websites

Example

Using public Wi-Fi for banking → attacker steals login details.

Q2: Vulnerability (Short Exam Enhancement)

Definition

A **vulnerability** is a weakness in a system that attackers can exploit.

Key Points (Important for Exams)

- Vulnerability ≠ Attack
- It is a **weak point**
- Attack happens when it is exploited

Types (Quick Revision Table)

Type	Example	Impact
Software	SQL Injection	Data theft
Hardware	Keylogger	Data capture
Network	Open ports	Unauthorized access
Configuration	Default password	Easy hacking

Type	Example	Impact
Human	Phishing	Account compromise

✅ Q3: Cyber Security Layers (Short Memory Trick)

👉 Remember: “PP EAD I OH”

Layer Meaning

- P Physical
 - P Perimeter
 - E Endpoint
 - A Application
 - D Data
 - I Identity
 - O Operational
 - H Human
-

✅ Q4: Cyber Stalking (Extra Points for Marks)

📌 Key Features

- Repeated harassment
- Fake accounts
- Monitoring activity
- Threat messages

📌 Detection Tip

- Check login activity
 - Look for unknown devices
 - Save evidence
-

✅ Q5: Malware vs Phishing vs MITM (Quick Comparison)

Feature	Malware	Phishing	MITM
Type	Software attack	Social engineering	Network attack
Target	System	Human	Communication
Goal	Damage system	Steal info	Intercept data
Example	Virus	Fake email	Public Wi-Fi hack

✅ Q6: Cyber Security Policies (Easy Structure)

👉 Write in exam like this:

1. Access Control Policy → Who can access
 2. Password Policy → Strong passwords
 3. Data Protection Policy → Encryption, backup
 4. Network Security Policy → Firewalls, VPN
 5. Acceptable Use Policy → Proper usage
 6. Incident Response Policy → Handle attacks
 7. Backup Policy → Data recovery
-

✅ Q7: Cyber Stalking (Identification + Detection Shortcut)

📌 Identify

- Repeated messages
- Fake profiles
- Threats

📌 Detect

- Login history
 - Message pattern
 - IP tracking
-

✅ Q8: Cyber Crimes (Easy Classification Trick)

👉 “IPOGFS”

Letter Type

I	Individual
P	Property
O	Organization
G	Government
F	Financial
S	Society

✅ Q10: Threat vs Vulnerability (Important Concept)

Term	Meaning
Threat	Possible danger
Vulnerability	Weakness
Attack	Exploitation

📌 Example:

Weak password (vulnerability) + hacker (threat) = breach (attack)

✅ Final Exam Tips 📖

- Always write:
 - Definition
 - Explanation
 - Example
 - Prevention
- Draw diagram for **MITM**
- Use tables for **differences**

- Add keywords: *Confidentiality, Integrity, Availability*

Q9. Software Attacks & Hardware Attacks

◆ 1) Software Attacks

✓ Definition:

Software attacks are malicious activities that target software systems such as operating systems, applications, or databases to gain unauthorized access, steal data, or disrupt services.

✓ Detailed Explanation:

- These attacks are carried out using malicious programs or code.
 - They exploit software bugs, vulnerabilities, or weak configurations.
 - Can be performed remotely through the internet or network.
 - Mainly affect **Confidentiality, Integrity, and Availability (CIA Triad)**.
-

✓ Characteristics:

- No physical access required
 - Can be executed remotely
 - Uses malware or exploit scripts
 - Targets software weaknesses
-

✓ Types of Software Attacks:

- Virus
 - Worm
 - Trojan Horse
 - Ransomware
 - Spyware
 - SQL Injection
 - Phishing
-

✓ Examples:

- **Virus** → Corrupts files and spreads to other systems
 - **Ransomware** → Locks data and demands payment
 - **SQL Injection** → Accesses database using malicious queries
 - **Trojan Horse** → Malicious code hidden inside useful software
-

✓ **Effects:**

- Data theft or loss
 - System crash or slowdown
 - Unauthorized system access
 - Financial loss
 - Privacy violation
-

✓ **Prevention:**

- Use updated antivirus software
 - Install security patches regularly
 - Avoid unknown links/downloads
 - Use firewalls and secure coding
-

◆ **2) Hardware Attacks**

✓ **Definition:**

Hardware attacks are attacks that target physical components of a computer system such as CPU, memory, hard disk, or network devices to steal, modify, or damage data.

✓ **Detailed Explanation:**

- Often require physical or close access to the system
 - Attackers may install malicious hardware devices
 - Difficult to detect using software tools
 - Can cause permanent damage to system or data
-

✓ **Characteristics:**

- Requires physical interaction
- Hard to detect remotely

- Can bypass software security
 - May cause permanent damage
-

✓ Types of Hardware Attacks:

- Hardware keylogger
 - Chip-level tampering
 - Side-channel attacks (power/timing analysis)
 - Device theft or replacement
 - Firmware attacks
-

✓ Examples:

- **Hardware Keylogger** → Captures keystrokes secretly
 - **Hard Disk Theft** → Stealing storage to access data
 - **Chip Tampering** → Modifying processor or memory chips
 - **Side-channel attack** → Extracting data via power usage patterns
-

✓ Effects:

- Permanent data loss
 - Unauthorized access to sensitive data
 - System failure or malfunction
 - Physical damage to hardware
-

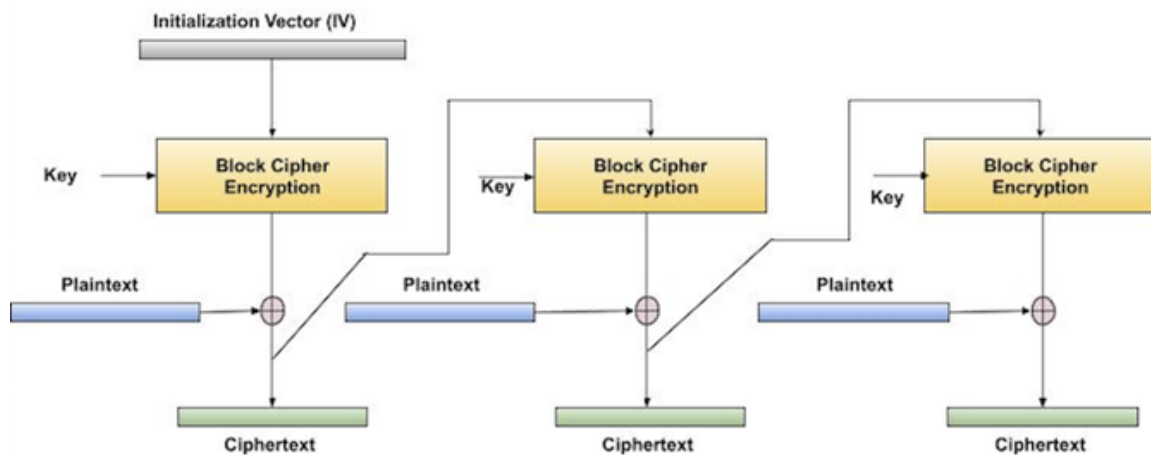
✓ Prevention:

- Use physical security (locks, CCTV)
 - Secure server rooms
 - Regular hardware inspection
 - Use hardware authentication methods
-

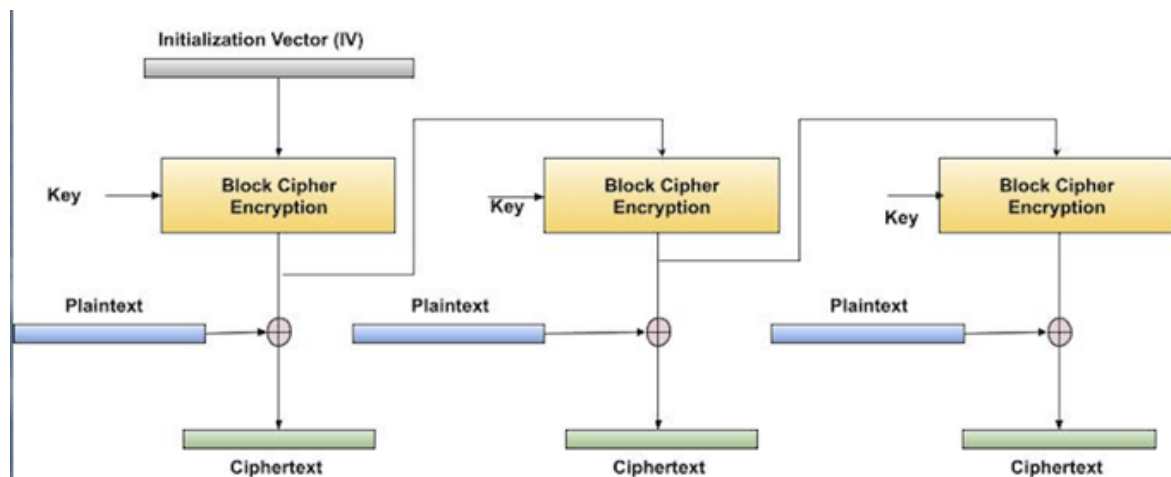
◆ 3) Difference Between Software & Hardware Attacks

Feature	Software Attacks	Hardware Attacks
Target	Software systems	Physical components

Feature	Software Attacks	Hardware Attacks
Access	Remote possible	Physical access needed
Detection	Easier using tools	Difficult to detect
Tools	Malware, exploits	Physical devices
Impact	Data/system damage	Permanent hardware damage



Cipher Feedback (CFB) Mode Encryption



Output Feedback (OFB) Mode Encryption